STATISTICS

Regression and Correlation Analysis PART 1

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Rev: 11.12.2024

Dependent and Independent Variables

The variables can be classified into two groups:

- Independent Variables
- Dependent Variables

Generally, we use X to denote independent variables and Y to denote dependent variables.

Examples:

- The working times (X) and exam grades (Y) of the students
- The incomes (X) and spending of people (Y)
- The growth of a plant (Y) and temperature (X)

Deterministic and Stochastic Models

The relations between variables can be classified as:

i. deterministic relationship

ii. stochastic (or semi-deterministic) relationship

For example, consider:

$X \longrightarrow$ The working times of the students

$Y \longrightarrow$ The grades of the students

if we know exactly the relationship between X and Y, such as:

$$Y=1.5X$$

There is no error term in this estimation.

deterministic model

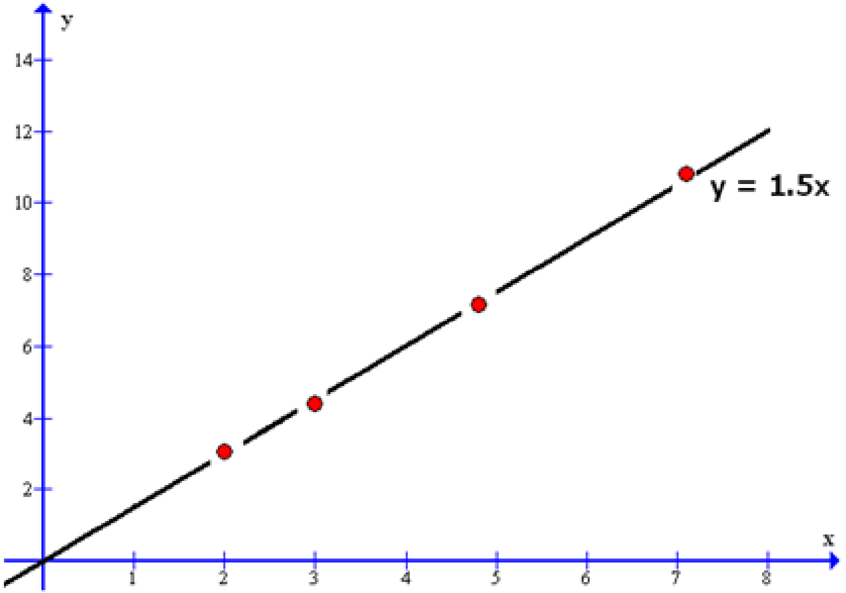
If the grades are also dependent on unknown random phenomena, then we should consider a random error:

$$Y=1.5X + Random\ Error$$

Then, Y becomes a random variable.

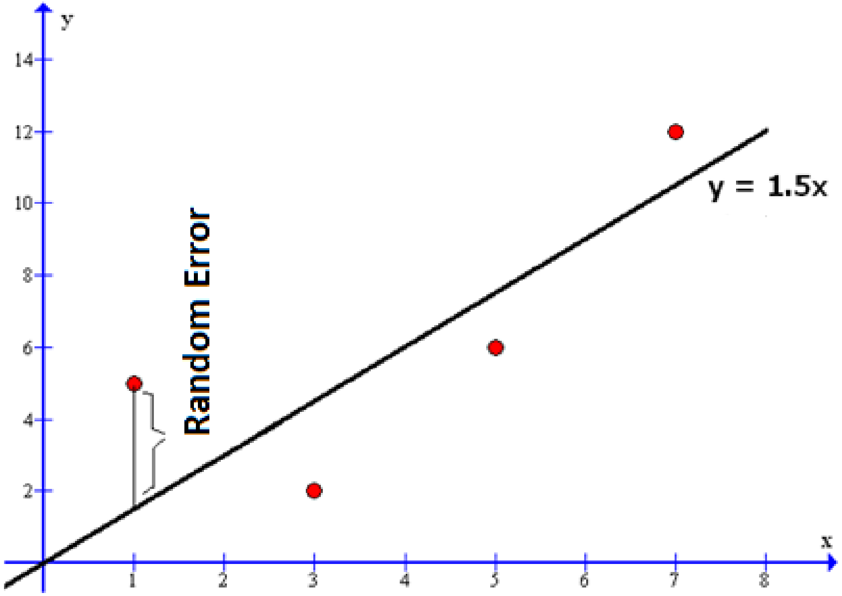
stochastic model

Deterministic and Stochastic Models



Deterministic Model:

$Y=1.5X$



Stochastic Model:

$Y=1.5X$ + Random Error

Regression and Correlation Analysis

Regression is the analysis of the stochastic (or semi-deterministic) relationship between two or more variables by using a **mathematical function**.

Correlation analysis deals with the **order** and **direction** of the relationship between these variables.

Consider the following data:

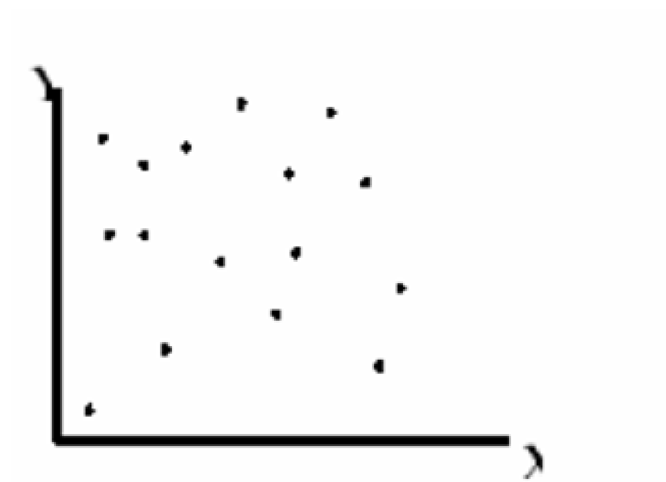
X	x_1	x_2	....	x_n
Y	y_1	y_2	....	y_n

Scatter Diagrams (Direction of the Relationship)

(a) The trend is linear with (+) direction

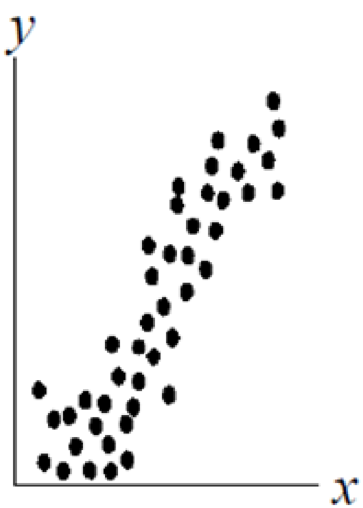
(b) The trend is linear with (-) direction

(c) The trend is non-linear

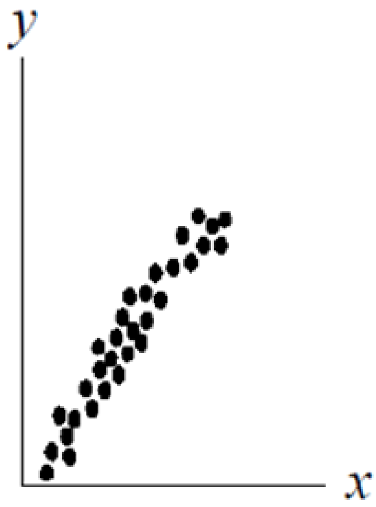


(d) There is no relationship

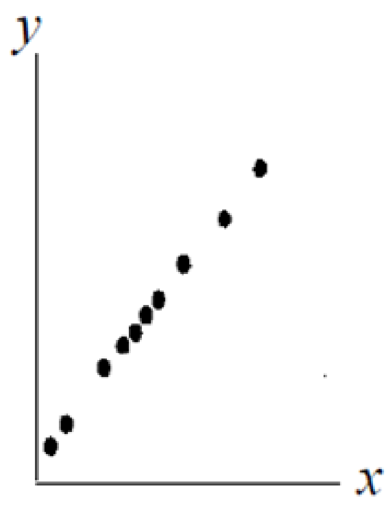
Scatter Diagrams (Order of the Relationship)



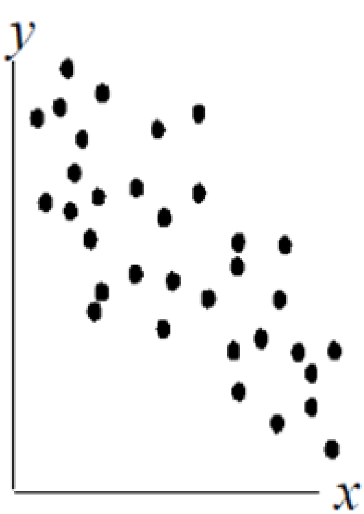
(a) Positive



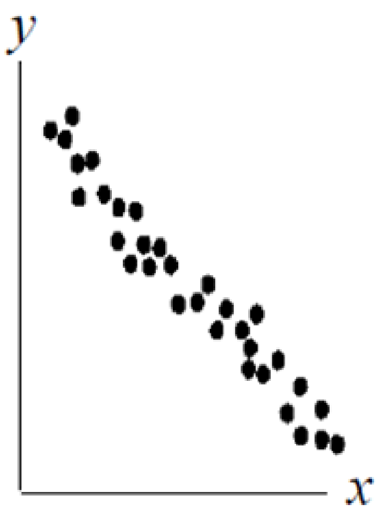
(b) Strong Positive



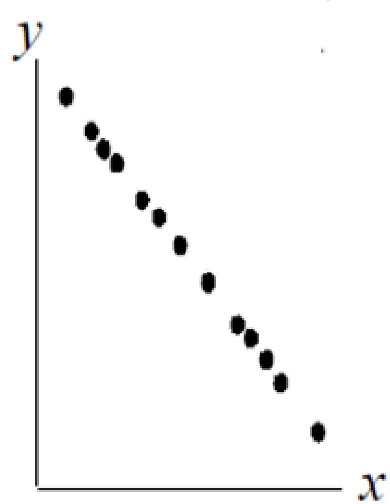
(c) Perfect Positive



(d) **Negative**



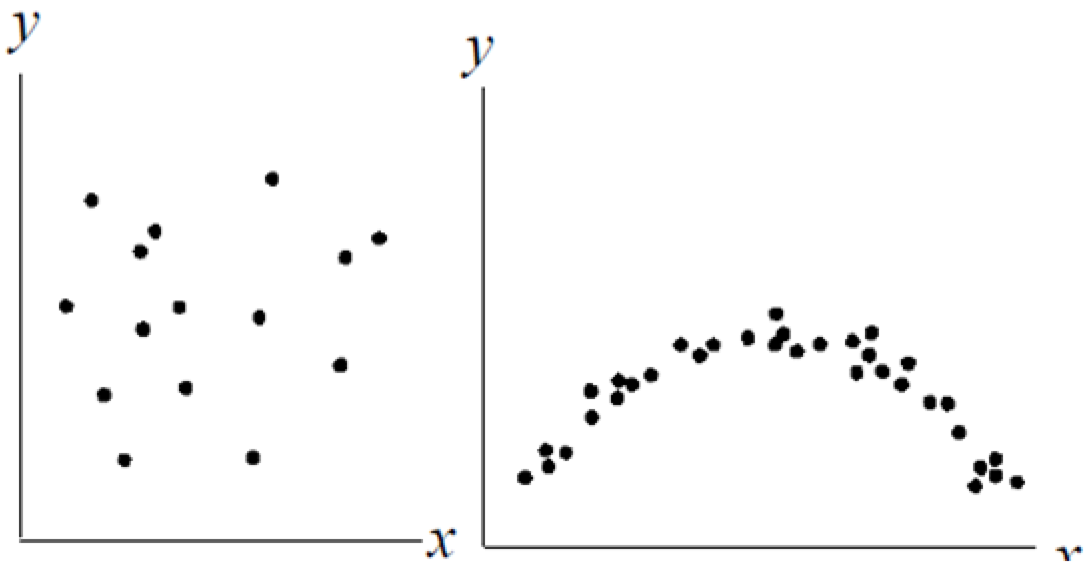
(e) **Strong Negative**



(f) **Perfect Negative**

Positive
and
Negative
Correlation

Scatter Diagrams (Order of the Relationship)



(g) No Correlation

(h) Non-linear Strong Relationship

The Relationship Between Variables

<u>Linear</u>

$$Y = \beta_0 + \beta_1 X$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

<u>Non-linear</u>

Non-linear relation:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2$$

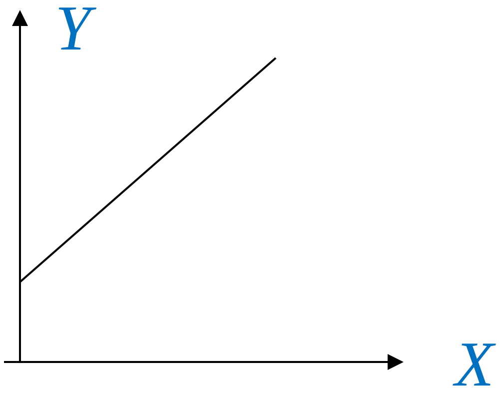
Logarithmical relation:

$$Y = \beta_0 + \beta_1 \log X$$

Exponential relation:

$$Y = \beta_0 + \beta_1 e^X$$

The simplest relation between two variables is the linear relation with two variables.



A general line equation is given by:

$$Y=\beta_0+\beta_1 X+\varepsilon$$

Here β_1 is the slope, and it shows the amount of change in Y according to the 1-unit change in X.

ε is the error.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

The above linear equation belongs to a population. To determine the coefficients β_0 and β_1, we first choose a sample from the population. Afterward, we determine the following line for the sample by using the least square method:

$$\hat{Y} = b_0 + b_1 X + e$$

The values of b_0 and b_1, which minimize the error term e, are estimators of β_0 and β_1.

Notation

Population Parameter	Sample statistic
β_0	b_0
β_1	b_1
$Y = \beta_0 + \beta_1 X + \varepsilon$	$\hat{Y} = b_0 + b_1 X$

LEAST SQUARE METHOD (LSM)

X	X_1	X_2	....	X_n
Y	Y_1	Y_2	....	Y_n

To determine the line $Y = b_0 + b_1 X + e$, more explicitly b_0 and b_1, that best represents the given points (X_i, Y_i), we minimize the error terms e:

$$Y = b_0 + b_1 X + e \Rightarrow e = Y - b_0 - b_1 X$$

If we substitute all (X_i, Y_i) point into the line equation and find the total sum of squares:

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right)^2$$

Here, X and Y are compact forms of X_i and Y_i, respectively.

The values of b_0 and b_1, which minimize the above expression, are estimators of β_0 and β_1.

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right)^2$$

We can find the value of the estimators by using the derivation of the above expression for b_0 and b_1, then equalizing them to 0.

Derivation with respect to $b_{0;}$

$$\frac{\partial}{\partial b_0} \sum_{i=1}^{n} e_i^2 = \frac{\partial}{\partial b_0} \sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right)^2$$

$$= -2 \sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right)$$

Derivation with respect to b_1;

$$\frac{\partial}{\partial b_1} \sum_{i=1}^{n} e_i^2 = \frac{\partial}{\partial b_1} \sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right)^2$$

$$= -2 \sum_{i=1}^{n} \mathbf{X} \left(\mathbf{Y} - \mathbf{b_0} - \mathbf{b_1 X}\right)$$

Equalizing them to 0;

$$-2 \sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right) = 0$$

$$\sum_{i=1}^{n} \left(Y - b_0 - b_1 X\right) = 0$$

$$-2 . \sum_{i=1}^{n} X . \left(Y - b_0 - b_1 X\right) = 0$$

$$\sum_{i=1}^{n} X . \left(Y - b_0 - b_1 X\right) = 0$$

$$-2\sum_{i=1}^{n}\left(Y-b_0-b_1X\right)=0 \qquad -2.\sum_{i=1}^{n}X.\left(Y-b_0-b_1X\right)=0$$

$$\sum_{i=1}^{n}\left(Y-b_0-b_1X\right)=0 \qquad \sum_{i=1}^{n}X.\left(Y-b_0-b_1X\right)=0$$

By expansion them, we obtain

$$\sum Y - n.b_0 - b_1\sum X = 0 \qquad \sum XY - b_0\sum X - b_1\sum X^2 = 0$$

These equations are called the normal equations of a line. We can find estimators by solving this system.

$$\left.\begin{aligned}\sum Y &= n.b_0 + b_1\sum X\\ \sum XY &= b_0\sum X + b_1\sum X^2\end{aligned}\right\} \quad b_1 = \frac{\sum XY - \frac{(\sum X).(\sum Y)}{n}}{\sum X^2 - \frac{(\sum X)^2}{n}}$$

$$b_0 = \bar{Y} - b_1\bar{X}$$

Here, $\bar{X}$ and $\bar{Y}$ are the means of X and Y, respectively.

alternatively, the system can solve by using Cramer's Rule:

$$\sum Y = n.b_0 + b_1 \sum X$$

$$\sum XY = b_0 \sum X + b_1 \sum X^2$$

$$b_0 = \frac{\begin{vmatrix} \sum Y & \sum X \\ \sum XY & \sum X^2 \end{vmatrix}}{\begin{vmatrix} n & \sum X \\ \sum X & \sum X^2 \end{vmatrix}} = \frac{\sum Y \sum X^2 - \sum X \sum XY}{n \sum X^2 - (\sum X)^2}$$

$$b_1 = \frac{\begin{vmatrix} n & \sum Y \\ \sum X & \sum XY \end{vmatrix}}{\begin{vmatrix} n & \sum X \\ \sum X & \sum X^2 \end{vmatrix}} = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

The Basic Assumptions of Regression Analysis

$$\boxed{Y = \beta_0 + \beta_1 x + \varepsilon}$$

- For given each x_i , the mean of the error term is zero, namely,

$$E(\varepsilon_i) = 0$$

- For given each x_i, the variance of errors are same, namely,

$$Var(\varepsilon_i) = \sigma_\varepsilon^2$$

- The errors are independent.

- For given each x_i, the errors are distributed normally.

Example 1: To research the relationship between age and maintenance spending of the trucks used for transportation in a factory, the data of 12 trucks were collected. Find the regression line and make the point estimation for the maintenance spending of a truck whose age is 4.

Here, age is the independent variable, and maintenance spending is the dependent variable. The regression line for the population is:

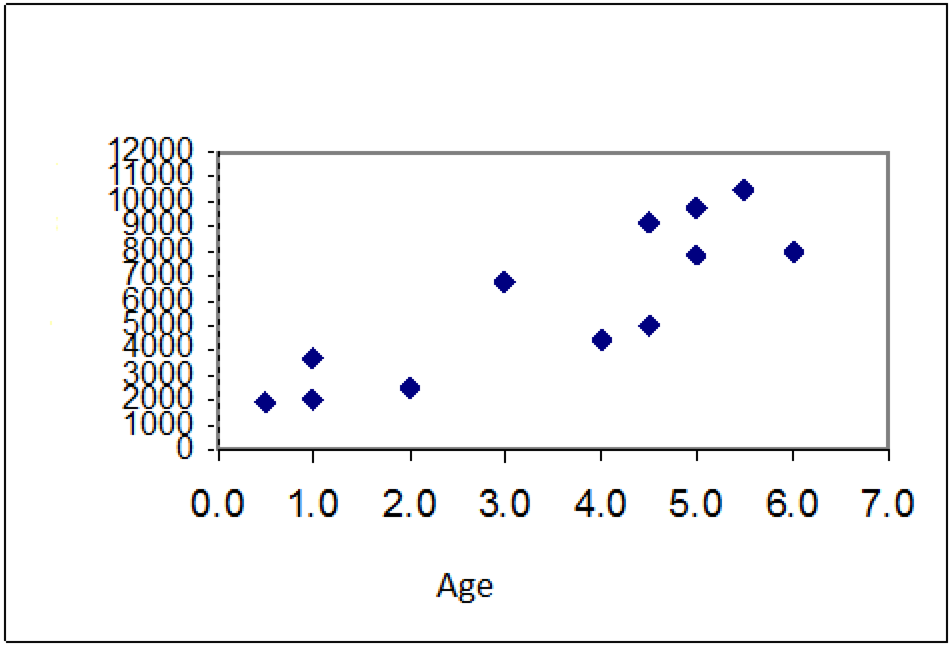
Maintenance Spending

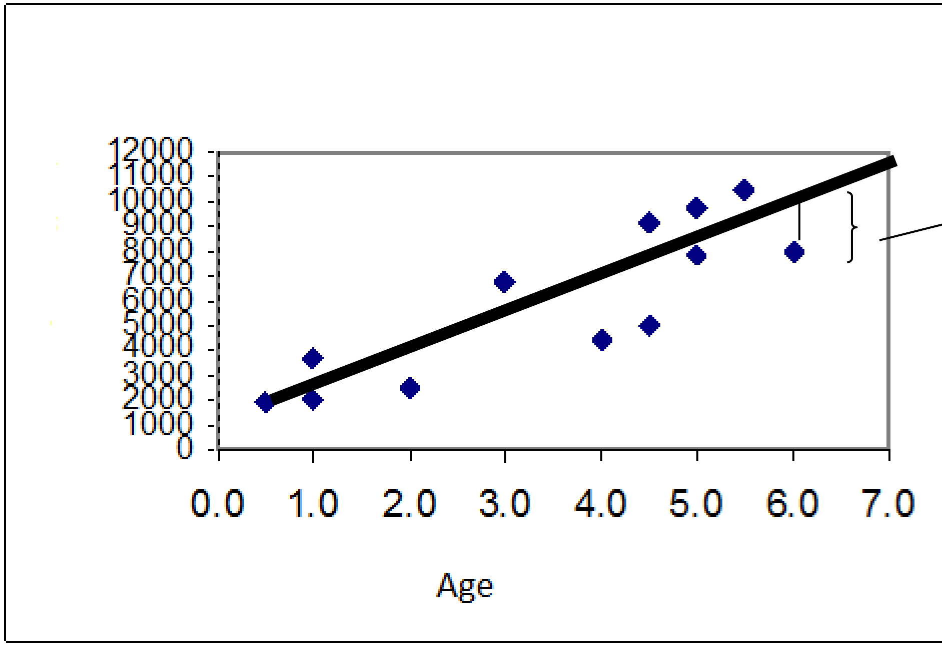
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Error Term

Age

Age (year)	Maintenance Spending
2.0	2500
4.5	9200
4.5	4950
4.0	4400
5.0	7900
5.5	10500
5.0	9700
0.5	1950
6.0	8000
1.0	2025
1.0	3700
3.0	6800





The error term **e** denotes the deviation of exact spending from the line.

To find the exact value of the population parameters β_0 and β_1, we should know the age and maintenance spending of **all** trucks in the factory. Generally, it is impossible, or it has a extremely high cost. For this reason, we try to estimate these parameters by using the sample.

Thus, the most appropriate line passing through the data points is:

$$\hat{Y} = b_0 + b_1 X$$

The estimator of the exact *Y*

	Age (Year) (x)	Maintenance spending (y)	x^2	y^2	xy
	2.0	2500	4	6250000	5000
	4.5	9200	20.25	84640000	41400
	4.5	4950	20.25	24502500	22275
	4.0	5500	16	30250000	22000
	5.0	7900	25	62410000	39500
	5.5	10500	30.25	110250000	57750
	5.0	9700	25	94090000	48500
	0.5	1950	0.25	3802500	975
	6.0	8000	36	64000000	48000
	1.0	2025	1	4100625	2025
	1.0	3700	1	13690000	3700
	3.0	6800	9	46240000	20400
Sum	42.0	72725.0	188.0	544225625.0	311525.0
Mean	3.5	6060.4			

$$\sum Y = n.b_0 + b_1 \sum X$$

$$\sum XY = b_0 \sum X + b_1 \sum X^2$$

$72725 = 12b_0 + 42b_1$

$311525 = 42b_0 + 188b_1$

$3.5*(72725 = 12b_0 + 42b_1)$

$311525 = 42b_0 + 188b_1$

$254537.5 = 42b_0 + 147b_1$

$- \quad 311525 = 42b_0 + 188b_1$

$-56988 = -41b_1$

$b_1 = 1390$

$72725 = 12b_0 + 42b_1$

$72725 = 12b_0 + 42*1390$

$b_0 = 1195$

The regression line for the sample is: $$\hat{Y} = 1195 + 1390X$$

The point estimation for the population regression line is: $Y = 1195 + 1390X$

By using this equation, we can make a point estimation for the maintenance spending of a truck whose age is known. For example, the maintenance spending of a truck whose age is 4 is:

$$\hat{Y} = 1195 + 1390X$$

$$\hat{Y} = 1195 + (1390)(4) = 6755$$

point estimation of Y when $X = 4$.

Example 2: The yearly sales of a company in 1993-1999 are given below. According to these values find the estimation of the regression equation.(x1000 units)

Years	Sales (Y)	X	XY	X^2
1993	15	1	15	1
1994	18	2	36	4
1995	25	3	75	9
1996	30	4	120	16
1997	40	5	200	25
1998	60	6	360	36
1999	82	7	574	49
$\sum$	**270**	**28**	**1380**	**140**

$$\sum Y_i = nb_0 + b_1 \sum X_i$$

$$\sum X_i Y_i = b_0 \sum X_i + b_1 \sum X_i^2$$

$$270 = 7b_0 + 28b_1$$

$$1380 = 28b_0 + 140b_1$$

$$1080 = 28b_0 + 112b_1$$

$$1380 = 28b_0 + 140b_1$$

$$300 = 28b_1 \Rightarrow b_1 = 300/28 = 10.7$$

$$b_1 = \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sum X^2 - \frac{(\sum X)^2}{n}} = \frac{1380 - \frac{(28)(270)}{7}}{140 - \frac{(28)^2}{7}} = 10.7$$

$$b_0 = \bar{Y} - b_1 \bar{X} = \frac{270}{7} - (10.7)\frac{28}{7} = -4.2$$

$$\hat{Y} = -4.2 + 10.7X$$

or by using Cramer's Rule;

$$b_0 = \frac{\sum Y \sum X^2 - \sum X \sum XY}{n\sum X^2 - (\sum X)^2} = \frac{270(140) - 28(1380)}{7(140) - (28)^2} = -\frac{30}{7}$$

$$b_1 = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = \frac{7(1380) - 28(270)}{7(140) - (28)^2} = \frac{75}{7}$$

$$\left.\right\} \quad Y = -\frac{30}{7} + \frac{75}{7}X$$

Another way of parameter estimation is the translation origin. In this case, the sum of *X* may become zero. Thus,

$$b_0 = \frac{\sum Y}{n} = \bar{Y} \qquad b_1 = \frac{\sum XY}{\sum X^2}$$

Yıllar	Y	X	XY	X^2	Y^2
1993	15	-3	-45	9	225
1994	18	-2	-36	4	324
1995	25	-1	-25	1	625
1996	30	0	0	0	900
1997	40	1	40	1	1600
1998	60	2	120	4	3600
1999	82	3	246	9	6724
	270	0	300	28	13998

$$b_0 = \frac{\sum Y}{n} = \bar{Y} = \frac{270}{7}$$

$$b_1 = \frac{\sum XY}{\sum X^2} = \frac{300}{28} = \frac{75}{7}$$

$$Y = \frac{270}{7} + \frac{75}{7}X$$

$$Y = -\frac{30}{7} + \frac{75}{7}x$$

a) What is the amount of sales in 2001?

Y2001=(-30/7)+(75/7) (9) = 92.1429

b) In which year, the sales are 100 units ?

100 = (-30/7)+(75/7)x

x = 9.7333

Year: 2001 Month: ? (0.7333)

For the translation origin, perform same example

$$Y = \frac{270}{7} + \frac{75}{7}X$$

a) Y = (270/7)+(75/7)(5) = 92.1429

b) 100== (270/7)+(75/7) x

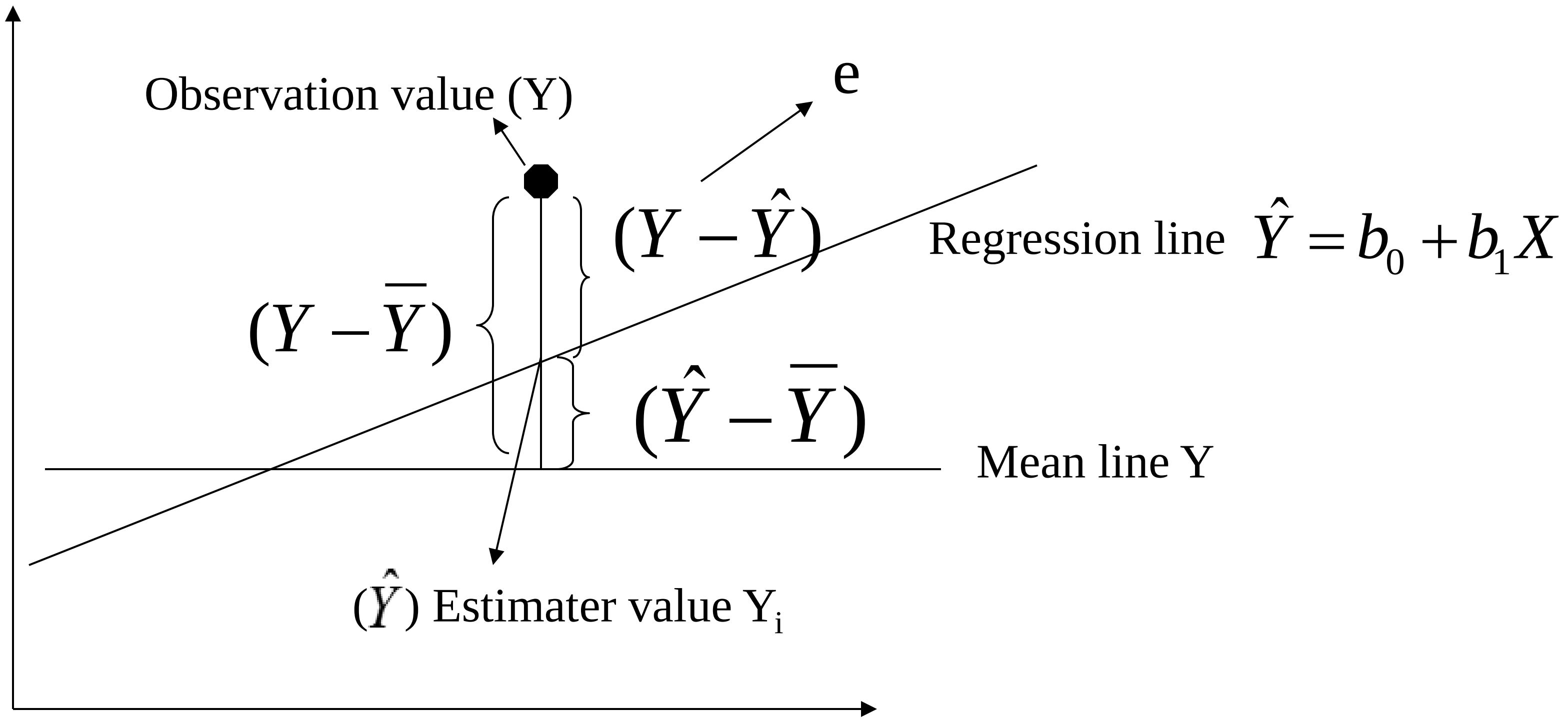
x=5.7333

Year: 2001 Month: ? (0.7333)

Sensitivity of the Estimated Regression

After we estimate the regression equation, we need to say how much the equation represents the relationship and how much the estimations are sensitive.

$$(Y - \hat{Y}) = (Y - \bar{Y}) - (\hat{Y} - \bar{Y})$$

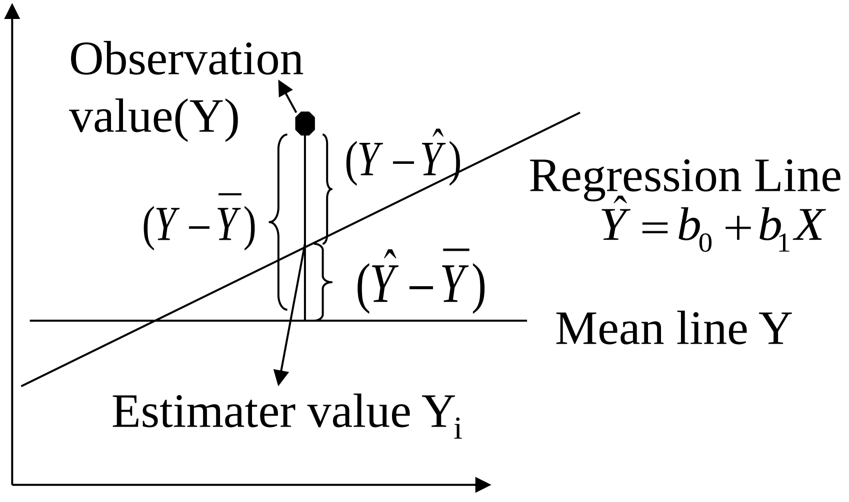


Coefficient of Determination (R^2)

If all the observations lie on the regression line, the sum of squares error is 0 and we say that fitting is perfect. According to this information, we can determine how much estimation is close to the real condition using the coefficient of determination, namely,

$$R^2 = \frac{\sum\left(\hat{Y}-\bar{Y}\right)^2}{\sum(Y-\bar{Y})^2} = \frac{\text{the sum of regression squares}}{\text{the sum of general squares}}$$

If R^2 approach 1, the fitting gets better. (generally $0<R^2<1$)



As seen from the figure, when observation values approach the regression line, the rate of the sum of regression squares to the sum of general squares (coefficient of determination) appraches 1.

Interpretation of Coefficient of Determination

$$R^2 = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2} = \frac{\text{the sum of regression squares}}{\text{the sum of general squares}}$$

The coefficient of determination shows

- how much percent of the dependent variable *Y* is dependent on the independent variable *X*

or

- how much percent of the variability in the data is explained by the regression model

or

- how well the independent variables explain the variability in the dependent variable.

Correlation Coefficient

ρ: Correlation coefficient of a population
r: Correlation coefficient of a sample

$$\rho = \frac{cov(X,Y)}{\sqrt{var(X)}\sqrt{var(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

$$r = \frac{S_{XY}}{S_X S_Y}$$

$$\rho = \frac{\frac{\sum(X-\mu_X)(Y-\mu_Y)}{n}}{\sqrt{\frac{\sum(x-\mu_X)^2}{n}}\sqrt{\frac{\sum(Y-\mu_Y)^2}{n}}}$$

$$= \frac{\sum(x-\mu_X)(Y-\mu_Y)}{\sqrt{\sum(x-\mu_X)^2}\sqrt{\sum(Y-\mu_Y)^2}}$$

$$r = \frac{\frac{\sum(X-\bar{X})(Y-\bar{Y})}{n-1}}{\sqrt{\frac{\sum(X-\bar{X})^2}{n-1}}\sqrt{\frac{\sum(Y-\bar{Y})^2}{n-1}}}$$

$$= \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2}\sqrt{\sum(Y-\bar{Y})^2}}$$

Correlation Coefficient

The correlation coefficient is a statistical measure that quantifies the strength and direction of a linear relationship between two variables (X and Y).

$$-1 \leq r \leq 1$$

$$-1 \leq \rho \leq 1$$

$r = -1$ → There is an exact negative correlation.

$r = 1$ → There is an exact positive correlation.

$r \approx -1$ → There is a strongly negative correlation.

$r \approx 1$ → There is a strongly positive correlation.

Correlation does not imply causation. Even if r is high, it does not mean one variable causes the other.

Correlation Coefficient

The square root of the coefficient of determination R^2 is correlation coefficient. The sign of the correlation coefficient is determined by the sign of the regression line slope.

$$r = \pm\sqrt{\frac{\sum(\hat{Y}-\bar{Y})^2}{\sum(Y-\bar{Y})^2}} \qquad \text{or} \qquad r = \frac{\sum xy}{\sqrt{(\sum x^2)(\sum y^2)}}$$

where;

$$\sum xy = \sum XY - \frac{(\sum X)(\sum Y)}{n} \qquad \sum x^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$\sum y^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

If we arrange it, we obtain;

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n\sum X^2 - (\sum X)^2\right]\left[n\sum Y^2 - (\sum Y)^2\right]}}$$